TESTING REPORT

PlaceIQ

Verification of the database, business rules, queries and web API: 91 of 91 automated tests passing

Fundamentals of Database Systems (FDS)

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1. Introduction

This report describes how PlaceIQ was verified: what was tested, how, in which environment, and with what result. All results were produced by the automated test suite in tests/, executed with pytest against a real PostgreSQL 18 database rebuilt from the project scripts. Every table of results in this document is generated from the suite's own output file (tests/results/results.json), not typed by hand.

1.1 Objectives

- Show that every business rule (BR-1 to BR-15) is enforced by the database, including when the application is bypassed.
- Show that the analytical queries and views return correct answers, using independent cross-checks.
- Show ACID behaviour, including correct isolation under real concurrency.
- Measure, rather than assume, the effect of indexes.
- Show that SQL injection is prevented and that the demonstration path is safely sandboxed.
- Record honestly every defect found during development and testing.

2. Test strategy

2.1 Test levels

Level	What is checked	Tests
Schema structure	The catalog contains exactly the designed objects: tables, keys, indexes, views, triggers, routines.	10
Declarative constraints	Each business rule enforced by the DDL rejects invalid data, and boundary values are accepted.	20
Triggers and procedures	Multi-table rules: eligibility, offer rules, the one-offer cascade, deadlines, expiry.	16
Views and queries	The five analytical queries and both views return provably correct answers.	9
Data invariants	The whole seeded dataset satisfies every rule and is reproducible.	9
Transactions & concurrency	Atomicity, isolation under two real concurrent sessions, deferred consistency at COMMIT.	4
Performance	Index effectiveness measured with EXPLAIN ANALYZE on 600,000 rows.	3
Security	Least-privilege role (DCL) and SQL-injection behaviour, sandboxing included.	9
REST API	Endpoints return correct data and surface database rule violations clearly.	11

2.2 Approach

- **Known starting state.** The session begins by rebuilding the database from 02_ddl.sql, 03_logic.sql and 04_data.sql. The seed is deterministic (setseed), so expected values such as '8 students match job 25' are stable.
- **Isolation between tests.** Each test receives a connection whose outer transaction is rolled back afterwards; nothing a test does persists.
- **Many violations per test.** Expected failures run inside a SAVEPOINT (helper expect_error) and assert the exact SQLSTATE and message, so one test can check several violations and continue.
- **Committing tests.** Tests that must commit (two-session concurrency, API writes) use a fixture that rebuilds the seed afterwards.
- **Independent oracles.** Query results are verified by a different formulation (for example relational division by double NOT EXISTS versus by counting) or by recomputing the expected value directly.
- **Negative and positive cases.** Each constraint test also checks that a legal boundary value is accepted (for example CGPA exactly 10, back-to-back interview slots).

2.3 Expected error codes

SQLSTATE	Meaning	Raised by
23514	check_violation	CHECK constraints; trigger RAISE … USING ERRCODE = 'check_violation'
23503	foreign_key_violation	Foreign keys (including the composite ones)
23505	unique_violation	UNIQUE / alternate keys
23P01	exclusion_violation	The interview double-booking EXCLUDE constraint
23001	restrict_violation	ON DELETE RESTRICT
23000	integrity_constraint_violation	Deferred totality trigger

SQLSTATE	Meaning	Raised by
25006	read_only_sql_transaction	Write attempted inside READ ONLY (injection lab)
42501	insufficient_privilege	placeiq_readonly role
55P03	lock_not_available	lock_timeout while waiting for the student row lock

3. Test environment

Component	Version / detail
Database	PostgreSQL 18.6 on x86_64-windows
Python	3.12.7
pytest	7.4.4
psycopg	3.2.9
Operating system	Windows 11
Database name / schema	placeiq / placeiq
Run started (UTC)	2026-10-29 11:54:31
Command	app\.venv\Scripts\python -m pytest tests -v

4. Results summary

Result

91 of 91 tests passed, 0 failed, 0 skipped. Total test time 10.3 s (excluding database rebuilds). The suite was run twice consecutively with identical results.

Table 1. *Results by module*

Module	Tests	Passed	Failed	Time (ms)
Schema structure	10	10	0	84
Declarative constraints	20	20	0	238
Triggers and procedures	16	16	0	286
Views and queries	9	9	0	226
Data invariants	9	9	0	1,473
Transactions & concurrency	4	4	0	2,018
Performance	3	3	0	2,675
Security	9	9	0	262
REST API	11	11	0	3,033

4.1 First run

The first execution of the suite reported 87 passed and 4 failed. All four were analysed before anything was changed: three were defects in the tests themselves (D-08, D-09) and one revealed that a claim in the demo script was inaccurate on PostgreSQL 18 (D-07). None was a defect in the schema or the business logic. After the corrections the suite passed 91 of 91, and passed again on an immediate re-run.

5. Requirements traceability

Every business rule is covered by at least one test; most by several (a negative case at the constraint level and a data-wide invariant).

Table 2. *Requirement → test cases*

Requirement	Test cases	Count
BR-2	TC-CON-03	1
BR-3	TC-CON-01	1
BR-4	TC-CON-04, TC-CON-05, TC-DAT-08	3
BR-5	TC-CON-06	1
BR-6	TC-CON-07	1
BR-7	TC-CON-08, TC-CON-09	2
BR-8	TC-CON-10, TC-DAT-04, TC-SCH-06	3
BR-9	TC-CON-12	1
BR-10	TC-CON-13	1
BR-11	TC-DAT-05, TC-TRG-01, TC-TRG-02, TC-TRG-03, TC-TRG-04, TC-TRG-05, TC-TRG-06, TC-TRG-07	8
BR-12	TC-DAT-03, TC-TRG-10, TC-TRG-12, TC-TRG-13, TC-TRG-14	5
BR-13	TC-CON-14	1
BR-14	TC-QRY-02, TC-TRG-11	2
BR-15	TC-CON-15	1
API	TC-API-01, TC-API-02, TC-API-03, TC-API-04, TC-API-05, TC-API-06, TC-API-07, TC-API-08, TC-API-09, TC-API-11	10
Atomicity	TC-TXN-01	1
Business logic	TC-SCH-09, TC-SCH-10	2
Consistency	TC-TXN-04	1
Data	TC-DAT-01, TC-DAT-02, TC-DAT-06, TC-DAT-09	4
Data load	TC-SCH-07	1
DCL	TC-SEC-01, TC-SEC-02, TC-SEC-03, TC-SEC-04	4
Defence in depth	TC-SEC-08	1
Design	TC-SCH-01	1
Domain	TC-CON-02, TC-CON-11, TC-CON-19, TC-CON-20	4
Domain integrity	TC-SCH-05	1
Entity integrity	TC-SCH-02	1
Function	TC-TRG-16	1
Indexing	TC-PRF-01, TC-PRF-02, TC-PRF-03	3
Injection	TC-SEC-05, TC-SEC-06, TC-SEC-07	3
Isolation	TC-TXN-02, TC-TXN-03	2
Offer rule	TC-DAT-07, TC-TRG-08, TC-TRG-09	3
Parameterisation	TC-SEC-09	1
Physical design	TC-SCH-04	1

Requirement	Test cases	Count
Procedure	TC-TRG-15	1
Q1	TC-QRY-04	1
Q2	TC-QRY-05, TC-QRY-06	2
Q3	TC-QRY-07	1
Q4	TC-QRY-08	1
Q5	TC-QRY-09	1
Queries	TC-QRY-01	1
Referential actions	TC-CON-16, TC-CON-17, TC-CON-18	3
Referential integrity	TC-SCH-03	1
UI	TC-API-10	1
Views	TC-QRY-03, TC-SCH-08	2

6. Detailed test cases

6.1 Schema structure

The catalog contains exactly the designed objects: tables, keys, indexes, views, triggers, routines.

ID	Req.	Test case	Result	ms
TC-SCH-01	Design	The schema contains exactly the 22 designed tables, plus the 2 analytics tables	PASS	8.2
TC-SCH-02	Entity integrity	Every table has a primary key	PASS	9.1
TC-SCH-03	Referential integrity	All 30 designed foreign keys exist, plus 2 on the analytics tables Includes the composite FKs used for disjointness, the sponsor category, the recruiter-company consistency rule and the interview aggregation.	PASS	5.4
TC-SCH-04	Physical design	Every foreign key is backed by an index that can find its rows PostgreSQL does not index FK columns automatically; an unindexed FK makes joins and ON DELETE checks scan the whole child table. Covered means: an index whose leading columns are exactly the FK columns, or a UNIQUE index on a leading subset of them (e.g. PK job_id covers FK (job_id, role_type)).	PASS	5.9
TC-SCH-05	Domain integrity	At least 40 CHECK constraints guard attribute domains	PASS	6
TC-SCH-06	BR-8	The interview double-booking EXCLUDE constraint exists	PASS	3.6
TC-SCH-07	Data load	Identity sequences were advanced past the explicitly inserted ids	PASS	24.7
TC-SCH-08	Views	The two operational views and the two analytics views exist	PASS	8.1
TC-SCH-09	Business logic	All 9 business-rule triggers are installed	PASS	7.4
TC-SCH-10	Business logic	The stored function and both procedures exist	PASS	6

6.2 Declarative constraints

Each business rule enforced by the DDL rejects invalid data, and boundary values are accepted.

ID	Req.	Test case	Result	ms
TC-CON-01	BR-3	CGPA outside 0-10 is rejected by a CHECK constraint	PASS	13.7
TC-CON-02	Domain	Malformed or upper-case e-mail addresses are rejected	PASS	4.1
TC-CON-03	BR-2	A duplicate roll number is rejected (UNIQUE alternate key)	PASS	11
TC-CON-04	BR-4	A full-time job cannot also be given an internship row (disjoint) The composite FK (job_id, role_type) only matches when role_type = 'INTERNSHIP'.	PASS	7.8
TC-CON-05	BR-4	A job without any subtype row fails the deferred totality check Inserting the job and its subtype in the same transaction passes.	PASS	17
TC-CON-06	BR-5	A recruiter cannot post a job for a company they do not work for	PASS	7.8
TC-CON-07	BR-6	A student cannot apply to the same job twice	PASS	22.4
TC-CON-08	BR-7	An interview in a round the job does not have is rejected (FK to weak entity)	PASS	18.3
TC-CON-09	BR-7	A student cannot be interviewed for a job they never applied to (aggregation FK)	PASS	8.9
TC-CON-10	BR-8	Overlapping interview slots for one student are rejected; back-to-back slots are allowed	PASS	11.5
TC-CON-11	Domain	An interview that ends before it starts is rejected	PASS	4.5
TC-CON-12	BR-9	A second offer on the same application is rejected	PASS	11.4

ID	Req.	Test case	Result	ms
TC-CON-13	BR-10	ACCEPTED or DECLINED offers must carry responded_at	PASS	7.4
TC-CON-14	BR-13	A company cannot claim a sponsor row typed ALUMNUS (category / union type)	PASS	5.5
TC-CON-15	BR-15	Deleting a student who holds an offer is blocked (CASCADE then RESTRICT)	PASS	28.6
TC-CON-16	Referential actions	Deleting a student with no offers cascades to all dependent rows	PASS	30.4
TC-CON-17	Referential actions	Deleting a recruiter nulls job_role.posted_by but keeps the job	PASS	13
TC-CON-18	Referential actions	A program that still has students cannot be deleted	PASS	7.2
TC-CON-19	Domain	Fixed pay above total CTC is rejected	PASS	3.6
TC-CON-20	Domain	A placement cycle that ends before it starts is rejected	PASS	3.8

6.3 Triggers and procedures

Multi-table rules: eligibility, offer rules, the one-offer cascade, deadlines, expiry.

ID	Req.	Test case	Result	ms
TC-TRG-01	BR-11	Applying with a CGPA below the job's cut-off is rejected by the trigger	PASS	10.6
TC-TRG-02	BR-11	A student who opted out of placements cannot apply	PASS	8.8
TC-TRG-03	BR-11	Applying after the application deadline is rejected	PASS	8.9
TC-TRG-04	BR-11	A student whose program is not eligible for the job cannot apply	PASS	18.4
TC-TRG-05	BR-11	A student with more active backlogs than the job allows cannot apply	PASS	19
TC-TRG-06	BR-11	A student already placed in the cycle cannot apply to another job in it	PASS	15.8
TC-TRG-07	BR-11	An eligible student's application is accepted with status APPLIED	PASS	20.3
TC-TRG-08	Offer rule	An offer cannot be made on a REJECTED or WITHDRAWN application	PASS	6.2
TC-TRG-09	Offer rule	Making an offer moves the application to OFFERED (AFTER INSERT trigger)	PASS	17.3
TC-TRG-10	BR-12	Accepting an offer declines other offers, withdraws open applications, frees interview slots and logs it Scenario: PGP62-084 holds a pending offer for job 19, gets a second offer for job 21, and is shortlisted with a scheduled interview for job 22.	PASS	29.8
TC-TRG-11	BR-14	After acceptance the derived status in v_placement_status becomes PLACED	PASS	42.1
TC-TRG-12	BR-12	An offer that was declined (manually or by the cascade) cannot later be accepted	PASS	24.4
TC-TRG-13	BR-12	A student placed in summer can still apply to and accept a final-placement offer	PASS	29.7
TC-TRG-14	BR-12	Accepting an offer after its response deadline is rejected	PASS	10.4
TC-TRG-15	Procedure	sp_expire_stale_offers() expires overdue pending offers and logs each one	PASS	19.1
TC-TRG-16	Function	fn_apply_for_job reports an unknown roll number clearly	PASS	4.9

6.4 Views and queries

The five analytical queries and both views return provably correct answers.

ID	Req.	Test case	Result	ms
TC-QRY-01	Queries	All five queries are present in 05_queries.sql and parse	PASS	0.1

ID	Req.	Test case	Result	ms
TC-QRY-02	BR-14	v_placement_status says PLACED exactly when the student has an accepted offer in that cycle	PASS	36.4
TC-QRY-03	Views	v_job_details has one row per job with exactly one of CTC / stipend	PASS	12.6
TC-QRY-04	Q1	Every company returned has >= 20 applicants and an offer rate below the campus average	PASS	22.2
TC-QRY-05	Q2	The talent match for job 25 returns the 8 expected students, led by DBA12-039	PASS	18.8
TC-QRY-06	Q2	Double NOT EXISTS gives the same answer as division-by-counting for every job Cross-checks the two classic SQL formulations of relational division.	PASS	33.9
TC-QRY-07	Q3	DENSE_RANK puts the program's highest CTC at rank 1 and above_avg_by equals CTC minus the program average	PASS	32
TC-QRY-08	Q4	The running total never decreases and ends at the program's number of final placements	PASS	16.2
TC-QRY-09	Q5	Every 'hidden gem' is unplaced, out-scored the selected students' average CGPA, and failed or missed a round	PASS	53.4

6.5 Data invariants

The whole seeded dataset satisfies every rule and is reproducible.

ID	Req.	Test case	Result	ms
TC-DAT-01	Data	The seed produces the documented volumes (reproducible with setseed)	PASS	17
TC-DAT-02	Data	Rebuilding the database twice produces byte-identical application and offer data	PASS	1401.3
TC-DAT-03	BR-12	No student holds more than one accepted offer in any cycle	PASS	9.1
TC-DAT-04	BR-8	No student has two overlapping interview slots	PASS	11.4
TC-DAT-05	BR-11	Every application meets its job's program, CGPA, backlog and deadline rules	PASS	10.5
TC-DAT-06	Data	Status timestamps never precede the event they follow	PASS	6.5
TC-DAT-07	Offer rule	Every application that received an offer has status OFFERED	PASS	5.8
TC-DAT-08	BR-4	Every job has exactly one subtype row (disjoint and total)	PASS	7.1
TC-DAT-09	Data	A student only sits round k of a job after clearing round k-1	PASS	4.3

6.6 Transactions & concurrency

Atomicity, isolation under two real concurrent sessions, deferred consistency at COMMIT.

ID	Req.	Test case	Result	ms
TC-TXN-01	Atomicity	Rolling back an acceptance undoes the entire trigger cascade	PASS	31.7
TC-TXN-02	Isolation	A second session accepting another offer for the same student waits on the student row lock	PASS	864.3
TC-TXN-03	Isolation	When the first session commits, the waiting session fails and the student ends with exactly one accepted offer Session B waits on the lock, then re-reads its offer, finds it DECLINED by A's trigger, and aborts.	PASS	908.9
TC-TXN-04	Consistency	COMMIT fails for a job without a subtype and leaves the database unchanged	PASS	213.3

6.7 Performance

Index effectiveness measured with EXPLAIN ANALYZE on 600,000 rows.

ID	Req.	Test case	Result	ms
TC-PRF-01	Indexing	A composite B-tree index on (job_id, status) replaces the sequential scan and is at least 10x faster	PASS	1013.6
TC-PRF-02	Indexing	A partial index on only the open rows is far smaller than the full composite index and still used	PASS	869
TC-PRF-03	Indexing	Without the leading column (job_id) the composite index cannot seek: the status-only query touches 50x more blocks Observed on PostgreSQL 18: the planner may still use the index, but only by reading all of it (or it falls back to a sequential scan), never by seeking.	PASS	792.6

6.8 Security

Least-privilege role (DCL) and SQL-injection behaviour, sandboxing included.

ID	Req.	Test case	Result	ms
TC-SEC-01	DCL	placeiq_readonly can SELECT the two tables it was granted	PASS	7.9
TC-SEC-02	DCL	placeiq_readonly is denied SELECT on tables it was not granted (offer)	PASS	8.5
TC-SEC-03	DCL	placeiq_readonly cannot INSERT, UPDATE, DELETE or DROP	PASS	32.9
TC-SEC-04	DCL	placeiq_readonly cannot call pg_read_file (unlike the postgres superuser)	PASS	1.8
TC-SEC-05	Injection	The deliberately unsafe search returns all 180 students for x' OR '1'='1	PASS	55.1
TC-SEC-06	Injection	The parameterised search returns 0 rows for the same payload	PASS	51.5
TC-SEC-07	Injection	A UNION payload on the unsafe path exposes phone numbers, showing the real risk	PASS	51.1
TC-SEC-08	Defence in depth	A DROP TABLE payload is refused by the read-only transaction and the table survives	PASS	47.7
TC-SEC-09	Parameterisation	An injection string passed to fn_apply_for_job is treated as a literal roll number	PASS	5.9

6.9 REST API

Endpoints return correct data and surface database rule violations clearly.

ID	Req.	Test case	Result	ms
TC-API-01	API	GET /api/health reports the placeiq database as reachable	PASS	95
TC-API-02	API	GET /api/summary returns the final-cycle KPIs (30 of 177 placed)	PASS	87.1
TC-API-03	API	Student search is case-insensitive and matches names	PASS	61.4
TC-API-04	API	An unknown roll number returns HTTP 404	PASS	47.3
TC-API-05	API	Jobs list has 25 jobs; job 19's funnel shows 70 applicants	PASS	150.7
TC-API-06	API	All five analytical queries run through the API in a read-only transaction	PASS	453.3
TC-API-07	API	A rejected application returns HTTP 409 with the SQLSTATE and the trigger that raised it	PASS	59.9
TC-API-08	API	Accepting an offer via the API returns the cascade the trigger performed	PASS	269.7
TC-API-09	API	Declining an offer that is not pending returns HTTP 409	PASS	255.1
TC-API-10	UI	The single-page UI and its assets are served	PASS	103.7
TC-API-11	API	POST /api/admin/reset rebuilds the seeded database	PASS	1449.6

7. Focus areas

7.1 Concurrency (TC-TXN-02, TC-TXN-03)

These tests open two real database connections. A second pending offer is committed for student PGP62-084, then:

1. Session A calls sp_accept_offer for the first offer and does not commit. It holds the student row lock.
2. Session B sets lock_timeout = 700 ms and calls sp_accept_offer for the second offer. It fails with SQLSTATE 55P03 (lock not available): B was blocked by A (TC-TXN-02).
3. In TC-TXN-03, B runs in a thread with no timeout. The test checks that B is still blocked after 0.6 s, then commits A.
4. B wakes, re-reads its offer under the lock, finds it DECLINED by A's cascade, and fails with 'only a PENDING offer can be accepted'.
5. Final check from a third connection: the student has exactly one accepted offer.

7.2 Performance (TC-PRF-01 to 03)

Table 3. *Measured on a 600,000-row archive, best of three EXPLAIN ANALYZE runs*

Measurement	Value
Sequential scan	60.839 ms
Composite B-tree (job_id, status)	0.025 ms (2,434× faster)
Partial index (open rows only)	0.018 ms
Full composite index size	4,488 KB
Partial index size	176 KB (26× smaller)
Blocks read: seek on job_id	3
Blocks read: status-only filter	4,383

Assertions are relative (at least 10× faster, at least 10× smaller, at least 50× more blocks) so that the tests are robust to machine speed.

7.3 Security (TC-SEC-01 to 09)

The least-privilege role can read only student and person; it is denied other tables, all writes, DROP, and pg_read_file. Through the API, the unsafe search returns all 180 students for x' OR '1'='1 and 180 phone numbers for a UNION payload, while the parameterised search returns 0 rows for the same input. A DROP TABLE payload is refused with SQLSTATE 25006 and the table is verified intact. Passing an injection string to fn_apply_for_job is treated as a literal roll number.

7.4 Query correctness

Q2 (relational division) is compared with the division-by-counting formulation for all 25 jobs; the sets are identical. Q1's rows are re-checked against an independently computed campus-wide offer rate; Q3's above_avg_by is recomputed; Q4's running total must be non-decreasing and end at the number of final placements per program; each Q5 row must be unplaced, out-score the selected students' average CGPA and show a failed or missed round.

8. Defect and issue log

All issues found during development and testing, including those in the tests themselves.

ID	Found by	Symptom	Root cause	Resolution	Status
D-01	Phase 2 smoke test	Inserting rows with explicit ids left the IDENTITY sequences at 1; the next default insert collided with an existing key.	Explicit ids do not advance a sequence.	setval() for every identity column at the end of 04_data.sql; verified by TC-SCH-07.	Fixed
D-02	Query review (Q4)	Some offers were accepted on 29 Oct, after the dataset's 'as of' date of 28 Oct.	Interview calendar started one day too late.	Final-placement interview days shifted one day earlier.	Fixed
D-03	Live demo dry run	After a pre-placement offer was auto-declined, that job's scheduled interview stayed booked.	Cascade only freed slots of WITHDRAWN applications, not of OFFERED ones whose offer was declined.	Cascade now deletes every SCHEDULED interview for the student's other jobs in the cycle; TC-TRG-10.	Fixed
D-04	Design review of concurrency	Two sessions accepting offers for the same student could deadlock.	Procedure locked the offer row before the student row.	sp_accept_offer locks the student row first (single lock order); TC-TXN-02/03.	Fixed
D-05	UI visual check	Dashboard panels in the same row were vertically misaligned.	A sibling-margin CSS rule applied inside grids.	Spacing applied only between top-level blocks.	Fixed
D-06	UI visual check (mobile)	Page overflowed horizontally at phone width.	CSS grid items default to min-width:auto.	minmax(0, 1fr) and min-width:0 on layout children.	Fixed
D-07	First test run, TC-PRF-03	PostgreSQL 18 used the (job_id, status) index for a status-only filter, contradicting the demo script.	The textbook rule is about seeking; PG 18 can read the whole index.	Test now compares blocks read (3 vs 4,383); demo wording corrected.	Fixed (documentation)
D-08	First test run, TC-DAT-01	A row created by one test persisted into later tests.	psycopg's conn.transaction() on an idle connection COMMITs; it is a savepoint only inside an open transaction.	The db fixture opens the outer transaction immediately.	Fixed (test harness)
D-09	First test run, TC-SCH-03/04	Expected 34 foreign keys (actual 30); composite FKs reported as unindexed although a UNIQUE index on their leading column covers them.	Wrong expectations in the tests.	Expected count corrected; coverage rule accepts a unique leading-column index.	Fixed (test)

9. Running the tests

```
# from the Project folder, with PostgreSQL running and the app venv created
app\.venv\Scripts\pip install pytest httpx
app\.venv\Scripts\python -m pytest tests -v                      # human-readable
app\.venv\Scripts\python -m pytest tests --junitxml=tests\results\junit.xml

# connection settings: PGHOST, PGPORT, PGDATABASE, PGUSER, PGPASSWORD
# psql location (for the rebuild fixture): PLACEIQ_PSQL
```

Outputs: tests/results/results.json (every test with id, requirement, outcome and duration), tests/results/junit.xml (for CI tools) and tests/results/performance.json (timings).

10. Conclusion and limitations

All 91 tests pass, covering every business rule, the correctness of all five queries, ACID behaviour under real concurrency, measured index performance and SQL-injection defence. Development testing found nine issues, all resolved and recorded above.

Not covered

- Automated browser tests of the UI: the UI was verified through the API tests, visual checks at desktop and mobile widths, and the scripted demo recording.
- Load and stress testing beyond the 600,000-row indexing experiment.
- Crash-recovery (durability) testing, which is a guarantee of the PostgreSQL engine rather than of this schema.